A quantum correlation length

Spin-1 singlet dimers and trimers: xi = 2.4901024

What was measured

The connected correlation of physical singlet projectors on longitudinal diameter pairs. Distances count cubic-centre spacings. The covering sum includes nonorthogonal quantum interference.

Geometry and exact finite-width method

The cylinder is infinite along x and periodic with 2 × 2 centres across y,z. Its full norm transfer has 4,096 states, separated into 64 invariant sectors. No tensor singular values are discarded.

Resolved exponential envelope

lambda_0 = 306.7333492693
lambda_dimer = -205.2828718653
xi = -1/log|lambda_dimer/lambda_0| = 2.4901024017
The negative eigenvalue gives alternating signs, not a power law.

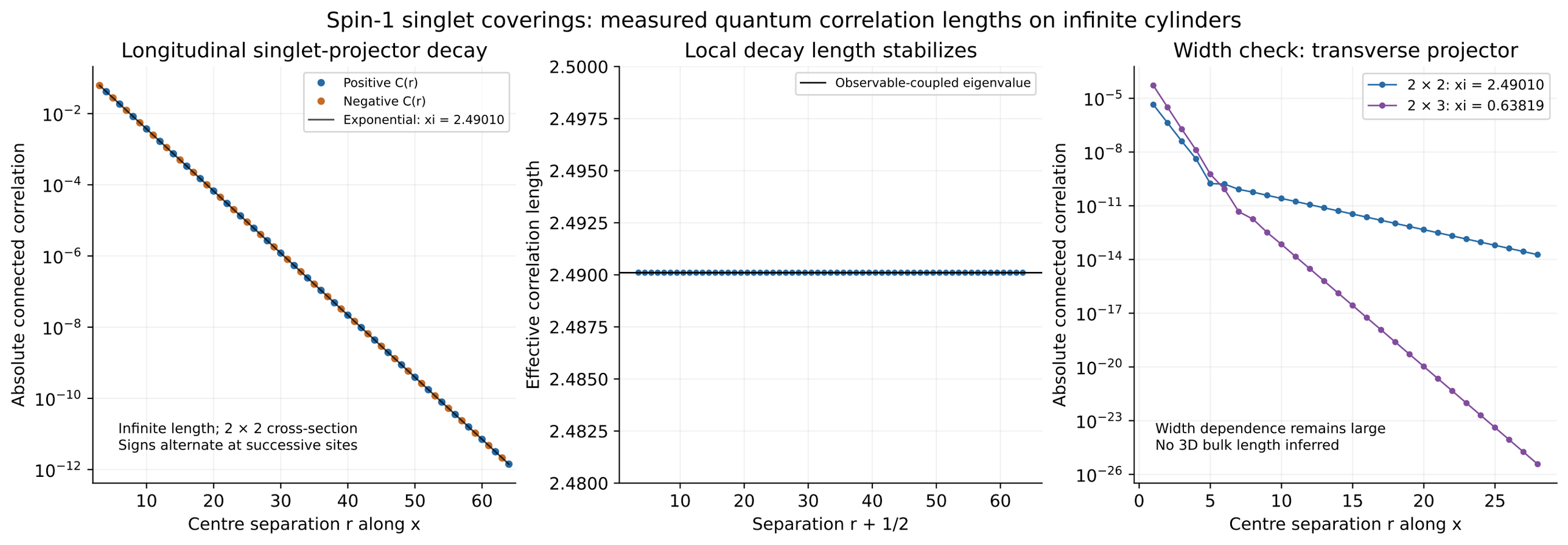
Independent checks

Physical projector insertions reproduce the exact 24-spin check. Two cylinder contractions of the same 36-spin torus agree. Direct correlation propagation, several fit windows and an 80-digit scalar calculation agree on the 2 × 2 decay length.

What remains unresolved

The same transverse projector gives xi = 0.6381923 on a 2 × 3 cylinder, versus 2.4901024 on 2 × 2. This large width dependence prevents a three-dimensional bulk extrapolation. These are computed quantum cylinder lengths, not a bulk claim.

Reproduce: python scripts/run_octahedral_spin1_cylinder.py
Precision check: python scripts/check_spin1_cylinder_precision.py


Spin-1 singlet coverings: measured quantum correlation lengths on infinite cylinders
Longitudinal singlet-projector decay
Positive C(r)
Negative C(r)
Exponential: xi = 2.49010
Infinite length; 2 × 2 cross-section
Signs alternate at successive sites
Absolute connected correlation
Centre separation r along x
Local decay length stabilizes
Observable-coupled eigenvalue
Effective correlation length
Separation r + 1/2
Width check: transverse projector
2 × 2: xi = 2.49010
2 × 3: xi = 0.63819
Width dependence remains large
No 3D bulk length inferred
Absolute connected correlation
Centre separation r along x